Chapter 2. PERMUTATIONS

2.1 What is a permutation?

2.1.1 Product of permutations
2.1.2 Inverse permutations

Today we will learn how to
- **multiply two permutations,**
- **find the inverse of a permutation.**

How does the product of permutations look like? What is it? No mystery here!

Let γ and σ be permutations in S_n. Recall that both are bijections from X_n to X_n, thus their **composition** $\gamma \circ \sigma$ is bijective, as well. So, $\gamma \circ \sigma$ is again a **permutation** in S_n. To find $\gamma \circ \sigma$, note that we first apply σ and then γ:

$$\boxed{(\gamma \circ \sigma)(k) = \gamma(\sigma(k)), \quad \text{for all } k \in X_n.}$$

We write $\gamma\sigma$ to denote $\gamma \circ \sigma$, and refer to it as the **product of the permutations γ and σ.**

It is way more practical to use the matrix-type notation to find the product of two permutations.

For instance, let us compute $\gamma\sigma$ for

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 3 & 5 & 1 \end{pmatrix}, \quad \gamma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 1 & 2 & 3 \end{pmatrix}. \qquad S_5$$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 2 & 4 & 3 & 5 & 1 \end{pmatrix}$$

$$\gamma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 5 & 4 & 1 & 2 & 3 \end{pmatrix}$$

$$\gamma\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 1 & 3 & 5 \end{pmatrix}$$

We say that **two permutations** σ **and** τ **in** S_n **commute** if $\sigma\tau = \tau\sigma$.

Example 2.4. Compute $\sigma\tau$ and $\tau\sigma$ for

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}, \qquad \tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix}.$$

$\underline{\sigma\tau}$

$$\tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 2 & 4 & 3 & 1 \end{pmatrix}$$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 3 & 4 & 1 & 2 \end{pmatrix}$$

$$\sigma\tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix}$$

$\underline{\tau\sigma}$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 3 & 4 & 1 & 2 \end{pmatrix}$$

$$\tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 2 & 4 & 3 & 1 \end{pmatrix}$$

$$\tau\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 2 & 4 \end{pmatrix}$$

Remark if σ, τ and μ are permutations in S_n then we always have that $(\sigma\tau)\mu = \sigma(\tau\mu)$, since the composition of maps is associative.

Definition 2.5. The **identity permutation** ε in S_n is defined as

$$\varepsilon = \begin{pmatrix} 1 & 2 & \dots & n \\ 1 & 2 & \dots & n \end{pmatrix}.$$

In other words, ε is just the identity map: $\varepsilon(k) = k$ for every $k \in X_n$. It is easy to verify that $\varepsilon\sigma = \sigma = \sigma\varepsilon$, for all $\sigma \in S_n$. So ε plays the role in S_n that the number 1 plays for multiplication of numbers.

Inverse of a permutation

Every $\sigma \in S_n$ is a bijective map $\sigma : X_n \to X_n$, and so there exists a unique map $\sigma^{-1} : X_n \to X_n$ (which is an element of S_n) called the **inverse** of σ such that $\sigma\big(\sigma^{-1}(k)\big) = k,\ \sigma^{-1}\big(\sigma(k)\big) = k$. These equations tell us that each of σ and σ^{-1} reverses the action of the other. Thus **the matrix expression of** σ^{-1} can be obtained from the matrix of σ by reading up:

$$\sigma = \begin{pmatrix} 1 & 2 & \dots & n \\ \uparrow & \uparrow & & \uparrow \\ \sigma(1) & \sigma(2) & \dots & \sigma(n) \end{pmatrix}$$

Example

the inverse permutation of $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 1 & 8 & 3 & 2 & 5 & 6 & 7 \end{pmatrix} \in S_8$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ 4 & 1 & 8 & 3 & 2 & 5 & 6 & 7 \end{pmatrix}$$

$$\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 5 & 4 & 1 & 6 & 7 & 8 & 3 \end{pmatrix}$$

Exercise

$$\sigma \sigma^{-1} = \varepsilon = \sigma^{-1} \sigma$$